

Chapter 9 — Occupational Health in Urology

- **Urology is physically and emotionally demanding** and practised in a potentially hazardous environment; occupational health affects performance, surgical outcomes and well-being.
- **Surgical culture devalues self-care** and rewards grit, efficiency and self-sacrifice, so attention to occupational health is countercultural — yet managing it optimises the surgeon's health, performance and quality of care.

PHYSICAL ERGONOMICS

Musculoskeletal Strain and Injury: The Importance of Surgical Ergonomics

- **Work-related musculoskeletal disorders (MSDs) affect 46–90% of urologists** during their careers (survey data).
- **Pain degrades performance:** reduced range of motion, strength, motor control and mental focus; increased irritability.
- **Site of MSD tracks the type of surgery:**
 - Open → neck, shoulder, low back.
 - Robotic/endoscopic → eye strain, neck pain, hand/wrist strain.
- **Non-modifiable risk factors:** younger age, inexperience, anthropometrics (short stature, small glove size, low grip strength), female sex, high body mass index (BMI), and medical conditions (arthritis, neuropathy, varicose veins, poor circulation).
- **Prevention targets modifiable factors** (Table 9.1): prolonged static postures, repetitive motions and awkward positions.
- **Environmental adjustment** — table height, monitor location/height, foot-pedal position — optimises posture; targets differ by technique (Figs. 9.1–9.4).
- **Strain reducers:** antifatigue mats, two-piece lead, lightweight headlamps, lightweight steep-declination loupes.
- **"Postural resets":** regularly check position, identify awkward postures, correct toward neutral.
- **Outside theatre:** regular exercise and strength training appear protective; daily 2-minute neck resistance exercises improved chronic neck discomfort in a trial of non-surgeons.
- **Intraoperative microbreaks/stretchers** reduced pain and improved subjective focus **without lengthening cases** (several small studies); a free web app prompts stretches at set intervals.
- **Exoskeletons (open) and robotic flexible ureteroscopy** improve ergonomics and pain but are not yet widely used.
- **Many urologists will need** medication, physiotherapy, massage, time off or case-mix change during their careers; prevention prolongs careers.

TABLE 9.1 MODIFIABLE ERGONOMIC RISKS IN UROLOGY

Domain	Risk	Mitigation
Intraoperative body mechanics	Repetitive movements; prolonged awkward or static postures; prolonged standing; poor endurance, muscle fatigue; insufficient recovery time	Monitors for gaze-down, no axial rotation; foot pedals aligned with torso toward target; table height matched to modality; position patient to align surgeon with target; ergonomics in the time-out; neutral postures; postural resets; stretches in and between cases; alternate sitting/standing; workspace eyewear; compression hosiery; antifatigue mats; exercise and strength training; alternate operating days with clinic/administration
Office body mechanics	Prolonged mouse use/typing; no adjustable desk or ergonomic chair; static positions	Neutral postures; top of screen at eye level; alternate sitting/standing; workspace eyewear; dictate rather than type; adjustable desk, ergonomic chair
Equipment type, location	Heavy lead; one-piece lead loading the shoulders; monitors on top of tower or lateral; headlamps; loupes	Lightest lead, removed between cases; two-piece lead or belted one-piece (weight to hips); reposition monitors; lightest loupes, steep declination (neck <20°) , set to personal working distance; lightest headlamp
Instruments/scopes	Heavy scopes; non-ergonomic scope positions; instruments hard to palm; staplers hard to fire one-handed	Lighter (e.g. disposable) scopes; robotic ureteroscopy; lobby industry to design for the full range of surgeon anthropometrics
Lifting and moving patients	Obese or uncooperative patients; lateral bed-to-trolley transfer	At least two people , coordinated lift; mechanical aids (especially obesity); pivot, never twist under load
Practice characteristics	High volume and acuity; long complex cases; 12-hour lists, elective surgery ≥5 days/week; unable to scale back for injury	Alternate complex and simple cases; mix modalities (open, laparoscopic, endourological); vary case length; sit or vary posture while the resident operates; alternate operating and non-operating days; urology-specific disability insurance

- **What the table is for:** every risk is paired with a controllable countermeasure — the lesson is that most ergonomic load is modifiable at the level of set-up, scheduling and equipment choice.
- **Pattern:** the same three fixes recur across domains — neutral posture, monitor at/below eye level for gaze-down, and variation (posture, case type, day type).
- **Trap:** one-piece lead is not the problem per se — it is **shoulder loading**; a belted one-piece transferring weight to the hips is acceptable.

FIGS. 9.1–9.4 ERGONOMIC TARGETS BY OPERATIVE MODALITY (COURTESY KRISTIN CHROUSER)

Modality	Table / working height	Posture targets
Open (9.1)	Target working area (often inside the patient) 2–4 inches (5–10 cm) below elbow height ; set for the tallest member, steps for others	Neck flexion <25° ; shoulders relaxed, torso–arm angle <30°; elbow 90–120°. Loupes with too short a working distance force neck flexion and elbow <90°
Laparoscopic (9.2)	Table at about the surgeon's pubic symphysis , so handles are at or just below elbow height	Monitor directly ahead (no trunk/neck rotation), top at eye level (gaze-down); torso–arm <30°; elbow 90–120°; forearm and wrist neutral between pronation and supination

Modality	Table / working height	Posture targets
Endourological (9.3; supine percutaneous nephrolithotomy shown; applies to ureteroscopy, transurethral resection of the prostate)	Target (urethral meatus, access sheath) between surgeon's pubic symphysis and umbilicus	Main endoscope monitor directly ahead, tops of monitors at eye level; elbow 90–120°
Robotic console (9.4)	Chair: adjustable height, lumbar support, lockable castors	Armrest so forearm is parallel to the floor, shoulders relaxed; clutch often, elbows close to torso; back flexion <15°; knee 90° at rest, max 120° reaching pedals; elbow 90–120°

- **Constant across all four:** elbow 90–120°, gaze-down monitor, relaxed shoulders — only the height reference changes (elbow for open, symphysis for laparoscopic, symphysis-to-umbilicus for endourology).
- **Trap:** open surgery sets height from the **elbow**, laparoscopy from the **instrument handles** — the working field in laparoscopy is inside the abdomen, the hands are not.

Presbyopia and Ergonomics in Urologists

- **About a decade into practice most urologists need near-vision correction** for surgery; existing wearers move to bifocals, trifocals or progressives.
- **General-purpose multifocals create ergonomic strain:** looking through a narrow lens segment drives neck extension and long static neck positions.
- **Prefer "workspace" lenses** correcting only near and intermediate vision; if standard progressives/bifocals are used, **lower the computer monitor**.

Computer Workstation Ergonomics

- **Desk work compounds MSD** (electronic records, messaging, video visits).
- **Set-up:** knees 90°, feet flat (or footrest); elbows 90°, wrists straight, arms supported.
- **Monitor top at or slightly below eye level** → gaze-down about 20°; screen **20–40 inches (50–100 cm)** away.
- **Eye rests and walking breaks;** a laptop is almost impossible to use ergonomically without a riser or separate keyboard (Video 9.1).

Concept — *Musculoskeletal injury is a cumulative-load problem: static, awkward postures sustained over years. Every intervention either restores neutral posture (set-up, eyewear, monitors) or interrupts the load (microbreaks, resets, varying cases), which is why cheap, simple measures work.*

OCCUPATIONAL EXPOSURES

Exposures to Infectious Diseases

- **Core controls:** hand hygiene, personal protective equipment (PPE), immunisation.
- **Centers for Disease Control and Prevention (CDC) recommends vaccination of health care workers** against hepatitis B, influenza, and measles–mumps–rubella.
- **Bloodborne pathogens** — human immunodeficiency virus (HIV), hepatitis B virus (HBV), hepatitis C virus (HCV) — are the concern because of intraoperative sharps injuries.
- **Reduce by** safe handling/disposal protocols and safety-engineered devices; institutions provide needlestick training, reporting systems and post-exposure protocols.

Surgical Smoke Inhalation

- **Hazards:** particulate matter, toxic gases, bioaerosols — a respiratory risk to the whole team.
- **Mitigation:** smoke evacuation, ventilation, respiratory protection, smoke-evacuating diathermy pencils.
- **Carbon dioxide laser ablation of genital condylomata** → **N95 mask + smoke evacuator with tip within 2 inches (5 cm)** of the site, to prevent human papillomavirus transmission to the team's airways.

Chemical Exposures

- **Sources:** disinfectants, anaesthetic agents, chemotherapeutics; mitigate with ventilation, PPE, safe handling and guidelines.
- **Intraoperative intravesical chemotherapy** → **two pairs of chemotherapy gloves**, impermeable disposable gown, face shield and goggles; a **closed-system drug-transfer device** adds protection against spills.

Ionising Radiation Exposure

- **Fluoroscopy** raises risk of cataract, infertility and cancer.
- **Protect:** lead apron, thyroid shield, lead glasses, lead gloves, plus dose-reduction techniques (Table 9.2) — keep exposure **as low as reasonably achievable (ALARA)**.
- **Institutions must supply dosimeters** and ongoing radiation-safety training; training correlates with lower occupational dose.
- **Pregnant surgeon: fetal dosimeter at waist level under the apron;** maternity or double-thickness lead; minimise participation in radiation cases where possible.

TABLE 9.2 STRATEGIES TO REDUCE RADIATION DOSE AND EXPOSURE TIME

Phase	Strategy
Machine settings and assistants	Low-dose default; pulsed fluoroscopy at lowest rate (e.g. 1 pulse/s); "last image hold" and "save and swap"; dedicated radiographer familiar with the case anatomy; standardised language with radiographers
Start of case	Collimate when the whole field is not needed; surgeon foot-pedal control (no confusion about when, spot vs continuous); maximise distance from the source, minimise distance to the image intensifier (less scatter)
During case	Limit screening time; spot images, not continuous ; minimise magnification (magnification increases dose)

- **Organising logic:** dose falls with less time, more distance, more shielding and a smaller, lower-rate beam — every row applies one of these.
- **Trap:** magnification feels harmless but **increases dose**; and the image intensifier should be **close** to the patient, the source **far**.

Intraoperative Distractions and Noise

- **Distractions and flow disruptions** impair focus, raise cognitive load and jeopardise safety — but some (urgent pages, instrument updates) are necessary.
- **Protect critical portions:** music paused, no non-emergency pages, no team breaks; state expectations at the briefing or time-out.
- **Noise** (equipment, conversation, alarms, movement) impairs communication, increases errors, task load and fatigue; prolonged high-intensity exposure causes hearing loss.
- **Measured theatre noise** exceeds recommendations for high-concentration work but does not usually reach hearing-damage levels.

- **Mitigate:** closed-loop communication, quiet critical phases, consider equipment noise in capital purchases; keep modifiable sound comfortable for the team and safe for the patient.

Concept — *Each exposure is controlled by the same hierarchy: remove or reduce the source, interpose a barrier, then protect the individual. Radiation (time, distance, shielding), smoke (evacuation, then N95) and chemotherapy (closed transfer, then double gloves) are the same principle applied to different hazards.*

PHYSICAL HEALTH OF SURGEONS

- **Heart disease, unintentional injury and suicide** are top-five causes of death at 25–54 years and top-ten at 55–64; each can be aggravated by occupational exposure and culture.
- **Atherosclerotic cardiovascular disease (ASCVD) is the most threatening modifiable cause of death at 35–64.**
- **Chronic job stress** worsens cardiovascular and mental health risk; **sleep deprivation** raises risk of fatal car crash; **burnout** leads to depression, substance misuse and suicide.
- **Physicians see their own primary care provider less than non-physicians, surgeons least of all.** Surgeons who use primary care and do regular aerobic and strength exercise report higher quality of life and less burnout.
- **Why urologists avoid care:**
 - *Time constraints* — schedules crowd out appointments.
 - *Culture of self-sacrifice* — reluctance to burden colleagues with cover.
 - *Medical knowledge* — self-diagnosis and self-treatment delay care.
 - *Fear of judgement* — by colleagues or regulators, especially for mental health care.
- **Surgeons are a costly, long-trained asset;** neglect of fitness, nutrition, lipid screening/treatment and stress shortens careers.
- **Treat lipids and hypertension** to prevent myocardial infarction and stroke; resilience training offsets chronic stress; **age-appropriate cancer screening** finds disease at treatable stages.
- **Evidence-informed biomarkers** worth early attention: low-density lipoprotein cholesterol; visceral fat and muscle mass (DEXA with body composition); cardiorespiratory fitness (VO₂ max); insulin sensitivity (oral glucose tolerance test) — targets for preventing ASCVD, type 2 diabetes and cancer.
- **Nutrition** is key but judged beyond the chapter's scope because of controversy.

Physiologic Response to Surgery

- **During routine surgery:** heart rate ↑, heart-rate variability ↓, salivary cortisol ↑ — **back to baseline about 30 minutes after.**
- **Magnitude varies between surgeons,** possibly with negative affectivity and trait anxiety; the long-term effect of repeated exposure is unknown.

Sleep

- **National Sleep Foundation: 7–9 hours' sleep opportunity (age 25–64); 7–8 hours (≥65).**
- **Sleep hygiene:** no large meals, caffeine or alcohol before bed; fixed bed and rise times; dark, quiet, cool room; no screens in the bedroom; daytime activity.
- **Barriers:** workload, on-call, administrative frustration, worry about ill patients or adverse events.
- **Sleep deprivation degrades objective performance by at least 12%** (task time, economy of motion, errors). Acute vs chronic effects are incompletely studied; individuals and systems should remove barriers.

Fitness

- **CDC: 150 min moderate or 75 min vigorous exercise per week + muscle strengthening on 2 days**, covering all major muscle groups.
- **Muscle strength and cardiorespiratory fitness correlate with longevity**; target exercise and treatment at areas of ergonomic stress.

Concept — *The surgeon's commonest killers are modifiable and are amplified by the job; the failure is not of knowledge but of uptake, driven by time, culture and fear of judgement. The chapter's remedy is to treat the surgeon as a patient with the same screening and prevention offered to others.*

MENTAL AND EMOTIONAL HEALTH OF SURGEONS

- **Job-related stress is common** and underlies several distinct but interrelated conditions: burnout, compassion fatigue and others.

Burnout

- **Coined by Freudenberger (1980)**: occupational syndrome of **emotional exhaustion, depersonalisation and low sense of personal accomplishment**.
- **Personal effects**: damaged relationships, substance misuse, depression/anxiety, suicide risk. **Professional effects**: poorer care, lower patient satisfaction, higher turnover.
- **Prevalence**: medical providers 40–60%; **urology about 50%**; general working population 28%.
- **Risk factors in urologists**: more hours, more patient encounters per week, **mid-career rather than early career**; chronic MSD pain also raises burnout and shortens careers.
- **Future detection**: wearable physiological monitors (burnout raises blood pressure and heart rate); DNA methylation or hair cortisol as biomarkers.
- **No universal solution**; tailor to practice and person. Effective levers: more autonomy, efficient work processes, less administrative burden, peer/community support.
- **Faculty spending ≥20% of time on personally meaningful work have lower burnout**.
- **Institutions** should monitor burnout and weigh well-being in safety and financial plans; self-care must not displace system-level change.

The Traumatized Provider ("Second Victim Syndrome")

- **Leriche**: every surgeon carries "a small cemetery" of failures.
- **After an adverse event or error**: post-traumatic stress symptoms, anxiety, depression; guilt, shame, lost confidence; insomnia, anorexia, headache.
- **Terminology**: "second victim" can offend patients, families and providers; "traumatized caregiver" is an alternative.
- **Consequences**: higher burnout, substance misuse, early retirement and suicide.
- **Barriers to help**: embarrassment, shame, fear of judgement, licensure/credentialing implications of mental-health treatment.
- **Response**: training should teach recognition in self and others; institutions provide "**psychological first aid**", ongoing support and counselling. **Peer support** reduces distress and improves coping; programmes improve safety culture.

Psychological Stress and Management Strategies

- **Drivers**: organisational, environmental, technological, personal, team and patient factors; stress affects the surgeon's technical and non-technical performance **and that of the team**.

- **Two levels of defence:** prevent stressors (usually system-level) and build individual stress-management skills.
- **Stress Continuum Model** (US military): leaders identify each colleague's zone and deploy resources to return them to readiness; nearer "ill" → greater need for professional help (Fig. 9.5).

FIG. 9.5 STRESS CONTINUUM MODEL (MODIFIED FROM NASH, US MARINE CORPS AND NAVY, 2011)

Zone	Description	Symptoms	Coping
Ready (green)	Minimal stress, balanced, optimal	Relaxed, focused, productive, ethical, having fun	Exercise, sleep, diet, mindfulness, socialising, hobbies
Reacting (yellow)	Mild stress, stretched	Irritability, restlessness, poor concentration	Self-care, time management, realistic goals, delegate/eliminate tasks, relaxation (deep breathing)
Injured (orange)	Moderate, overwhelmed	Anxiety, mood swings, low purpose, substance misuse, burnout	Support of friends/family, reduce workload, time off, healthy coping (journaling, relaxation)
Ill (red)	Severe, in crisis	Panic, severe depression, substance dependence, social/occupational impairment	Mental-health professional, support networks, sick leave

- **Pattern:** coping escalates from self-managed (Ready/Reacting) → social and workload change (Injured) → professional care (Ill). **Trap:** burnout sits in the **Injured** zone, not Ill.
- **Cognitive behavioural therapy (CBT) and mindfulness increase resilience under stress** (trials). CBT: thoughts, emotions and behaviours are interconnected — identify and challenge distorted thoughts, replace with rational ones; deliverable by a professional or self-taught.
- **Dialectical behavioural therapy (DBT)** = CBT + mindfulness, emphasising **acceptance and validation of emotion** alongside behaviour change.
- **Both rest on recognising cognitive distortions** (Table 9.3). **Imposter syndrome** — persistent inability to believe success is deserved — is a distortion (personalisation) linked to burnout, commonest in trainees, manageable with CBT.
- **Mental practice** (cognitive rehearsal) and **relaxation (breathing)** reduce stress in surgical trainees; energy/attention management, goal setting and performance routines work in other high-stress fields.
- **Personal strategies:** supportive relationships; workload management (prioritise, delegate, realistic goals); mindfulness; exercise, nutrition, sleep; hobbies and non-work pursuits.

TABLE 9.3 COMMON COGNITIVE DISTORTIONS EXPERIENCED IN MEDICINE AND THERAPEUTIC STRATEGIES

Distortion	Description (example)	Tactic
Overgeneralisation	Broad conclusion from few cases (one bad outcome = incompetence)	Self-compassion (recall prior successes)
Catastrophising	Exaggerated negative outcome (rejected grant ruins career)	Double standard (many rejected researchers succeed)
Confirmation bias	Favouring evidence that fits prior belief (ignoring inconvenient trial data)	Cognitive defusion (entertain alternatives)
Labelling	Simplistic judgemental labels ("drug seeker", so pain not assessed)	Exploratory questions (is the label fair?)
Emotional reasoning	Feelings, not evidence, guide decisions (Medicaid vs "good" insurance)	Cognitive restructuring (are emotions clouding judgement?)
Depersonalisation	Detachment from or distorted perception of self (imposter syndrome)	Self-compassion (effort taken; everyone errs)

Distortion	Description (example)	Tactic
Minimisation	Downplaying significance (personal wellness unimportant)	Double standard (would you advise family so?)
Tunnel vision	Over-focus on one task or emotion (shouting over a missing instrument)	Exploratory questions (what matters most now?)

- **Tactics recur in pairs:** self-compassion (overgeneralisation, depersonalisation); double standard (catastrophising, minimisation); exploratory questions (labelling, tunnel vision).
- **Trap: imposter syndrome is a cognitive distortion**, not a personality trait — hence it responds to CBT.

Conclusion

- **Occupational health spans ergonomics, exposures, physical and mental health;** evidence-based individual strategies plus institutional policy protect surgeons, enhance patient safety and optimise outcomes.

Concept — *Stress is a continuum, not a binary: the same person moves between zones, and the response must escalate with the zone. Self-care handles the green-yellow end; the orange-red end needs workload change, peers and professionals, and institutions own the stressors that individuals cannot fix.*

KEY POINTS

- **Work-related musculoskeletal injury is common**, from isometric contraction, awkward postures and poor monitor positioning.
- **Short intraoperative stretch breaks** reduce strain and improve focus.
- **Minimise radiation:** lowest settings, least time, lead apron and thyroid collar.
- **Physical health:** exercise, diet, sleep hygiene, standard primary-care screening.
- **Stress:** CBT and mindfulness, peer support, institutional action on sources of frustration.